

Film Capacitors

EMI Suppression Capacitors (MKP)

Series/Type: B81123
Date: June 2026

Typical applications

- Y1 class for interference suppression
- "Line to ground" applications
- Double insulation

Climatic

- Max. operating temperature: 110 °C
- Climatic category (IEC 60068-1:2013): 40/110/56

Construction

- Dielectric: polypropylene (MKP)
- Internal series connection
- Plastic case (UL 94 V-0)
- Epoxy resin sealing (UL 94 V-0)

Features

- Self-healing properties
- RoHS-compatible
- AEC-Q200E compliant

Terminals

- Parallel wire leads, lead-free tinned
- Standard lead lengths: 6 – 1 mm
- Special lead lengths available on request

Marking

Manufacturer's logo, lot number, date code, rated capacitance (coded), cap. tolerance (code letter), rated AC voltage, series number, sub-class (Y1), dielectric code (MKP), climatic category, passive flammability category, approvals.

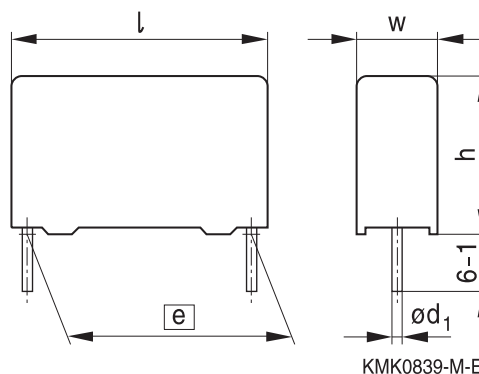
Delivery mode

Bulk (untaped)

Taped (Ammo pack or reel)

For taping details, refer to chapter "Taping and packing".

Dimensional drawing



Dimensions in mm

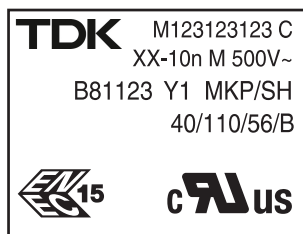
Lead spacing	Lead diameter d ₁
$e \pm 0.4$	
15, 22.5	0.8

EMI Suppression Capacitors (MKP)

B81123

Y1 / 500 V AC

Marking example (position of marks may vary):



Approvals

Approval marks	Standards	Certificate
	EN IEC 60384-14:2023	ENEC-05495
	UL 60384-14:2014, CSA E60348-14	E97863

Overview of available types

Lead spacing	15 mm	22.5 mm
C_R (μF)		
0.0010		
0.0015		
0.0022		
0.0033		
0.0047		
0.0056		
0.0068		
0.010		

Ordering codes and packing units

Lead spacing mm	C _R μF	Max. dimensions w × h × l mm	Ordering code (composition see below)	Ammo pack pcs./MOQ	Reel pcs./MOQ	Untaped pcs./MOQ
15	0.0010	5.0 × 10.5 × 18.0	B81123C1102M***	9360	10400	8000
	0.0015	6.0 × 11.0 × 18.0	B81123C1152M***	7680	8800	8000
	0.0022	7.0 × 12.5 × 18.0	B81123C1222M***	6640	7200	8000
	0.0033	8.5 × 14.5 × 18.0	B81123C1332M***	5440	5600	4000
	0.0047	9.0 × 17.5 × 18.0	B81123C1472M***	5120	5600	4000
22.5	0.0056	7.0 × 16.0 × 26.5	B81123C1562M***	4640	4800	5040
	0.0068	8.5 × 16.5 × 26.5	B81123C1682M***	3840	4000	4080
	0.010	10.5 × 18.5 × 26.5	B81123C1103M***	3120	3200	4320

MOQ = Minimum Order Quantity, consisting of 8 packing units.

Composition of ordering code

+ = Capacitance tolerance code:

M = ±20%

*** = Packaging code:

289 = Straight terminals, Ammo pack

189 = Straight terminals, Reel

003 = Straight terminals, untaped
(lead length 3.2 ±0.3 mm)

000 = Straight terminals, untaped
(lead length 6 – 1 mm)

Technical data

Reference standard: UL/IEC 60384-14:2023.

All data given at $T = 20\text{ °C}$, unless otherwise specified.

Max. operating temperature $T_{op,max}$	+110 °C	
Dissipation factor $\tan \delta$ (in 10^{-3}) at 20 °C (upper limit values)	at 1 kHz	1.0
	at 100 kHz	5.0
Insulation resistance R_{ins} or time constant $\tau = C_R \cdot R_{ins}$ at 20 °C, rel. humidity $\leq 65\%$ (minimum as-delivered values)	30 000 M Ω	
DC test voltage	4800 V, 2 s	
<i>The repetition of this DC voltage test may damage the capacitor. Special care must be taken in case of use several capacitors in a parallel configuration.</i>		
Passive flammability category	B	
Maximum continuous AC voltage V_{AC}	750 V (50/60 Hz)	
Rated AC voltage (UL 60384-14:2014)	500 V (50/60 Hz)	
Maximum continuous DC voltage V_{DC}	3000 V	
Temperature	$T_{op} \leq 110\text{ °C}$	$V_{op} = V_{AC}$ (continuously)
	$T_{op} \leq 110\text{ °C}$	$V_{op} = 1.7 \cdot V_{AC}$ (1000 h)
Humidity bias test	T = +85 °C $\pm$ 2 °C Relative humidity: 85% $\pm$ 2% V_R = 500 V AC, 50/60 Hz Test duration: 1000 hours	
Limit values after test	Capacitance change $ \Delta C/C $	$\leq 10\%$
	Dissipation factor change $\Delta \tan \delta$	$\leq 2 \cdot 10^{-2}$ (at 1 kHz)
	Insulation resistance R_{ins}	$\geq 200\text{ M}\Omega$

Pulse handling capability

"dV/dt" represents the maximum permissible voltage change per unit of time for non-sinusoidal voltages, expressed in V/μs.

"k₀" represents the maximum permissible pulse characteristic of the waveform applied to the capacitor, expressed in V²/μs.

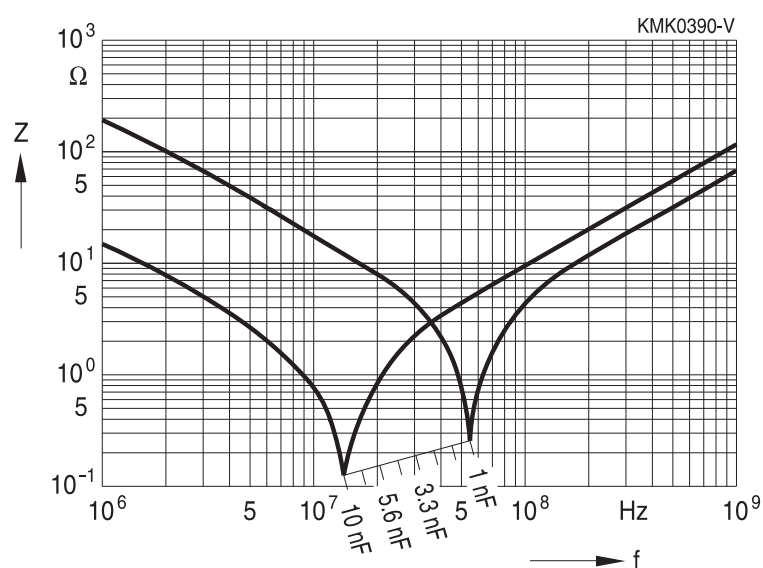
Note:

The values of dV/dt and k₀ provided below must not be exceeded in order to avoid damaging the capacitor.

dV/dt and k₀ values

Lead spacing	15 mm	22.5 mm
dV/dt in V/μs	3 000	1 000
k ₀ in V ² /μs	2 100 000	700 000

Impedance Z versus frequency f (typical values)



Testing and Standards

Test	Reference	Conditions of test	Performance requirements
Electrical parameters	IEC 60384-14:2023	Voltage Proof: Between terminals: 4000 V AC, 1 min Terminals and enclosure: 4000 V AC, 1 min Insulation resistance, R_{ins} Capacitance, C Dissipation factor, $\tan \delta$	Within specified limits
Robustness of terminations	IEC 60068-2-21:2006	Tensile strength (test Ua1) Wire diameter	Capacitance and $\tan \delta$ within specified limits
		Tensile force $0.5 < d_1 \leq 0.8 \text{ mm}$ 10 N $0.8 < d_1 \leq 1.25 \text{ mm}$ 20 N	
Resistance to soldering heat	IEC 60068-2-20:2008, test Tb, method 1A	Solder bath temperature at $260 \pm 5 \text{ }^\circ\text{C}$, immersion for 10 seconds	$\Delta C/C_0 \leq 5\%$ $\tan \delta$ within specified limits
Rapid change of temperature	IEC 60384-14:2023	T_A = lower category temperature T_B = upper category temperature Five cycles, duration $t = 30 \text{ min.}$	No visible damage $ \Delta C/C_0 \leq 5\%$ $\tan \delta$ within specified limits
Vibration	IEC 60384-14:2023	Test F_C : vibration sinusoidal Displacement: 0.75 mm Acceleration: 98 m/s^2 Frequency: 10 Hz ... 500 Hz Test duration: 3 orthogonal axes, 2 hours each axe	No visible damage
Bump	IEC 60384-14:2023	Test Eb: Total 4000 bumps with 400 m/s^2 mounted on PCB 6 ms duration	No visible damage $ \Delta C/C_0 \leq 5\%$ $\tan \delta$ within specified limits
Climatic sequence	IEC 60384-14:2023	Dry heat Tb / 16 h Damp heat cyclic, 1 st cycle $+55 \text{ }^\circ\text{C} / 24 \text{ h} / 95\% \dots 100\% \text{ RH}$ Cold Ta / 2 h Damp heat cyclic, 5 cycles $+55 \text{ }^\circ\text{C} / 24 \text{ h} / 95\% \dots 100\% \text{ RH}$	No visible damage $ \Delta C/C_0 \leq 5\%$ $ \Delta \tan \delta \leq 0.008$ for $C \leq 1 \text{ } \mu\text{F}$ $ \Delta \tan \delta \leq 0.005$ for $C > 1 \text{ } \mu\text{F}$ Voltage proof $R_{ins} \geq 50\%$ of initial limit
Humidity bias test	IEC 60384-14:2023	$85 \text{ }^\circ\text{C} / 85\% \text{ RH} / 500 \text{ V AC} / 1000 \text{ h}$	No visible damage $ \Delta C/C_0 \leq 10\%$ $ \Delta \tan \delta \leq 0.02$ at 1 KHz $R_{ins} \geq 200 \text{ M}\Omega$

Test	Reference	Conditions of test	Performance requirements
Damp heat, steady state	IEC 60384-14:2023	Test without voltage 40 °C / 93% RH / 56 days	No visible damage $ \Delta C/C_0 \leq 5\%$ $ \Delta \tan \delta \leq 0.008$ for $C \leq 1 \mu\text{F}$ $ \Delta \tan \delta \leq 0.005$ for $C > 1 \mu\text{F}$ Voltage proof $R_{\text{ins}} \geq 50\%$ of initial limit
		Test with rated AC voltage 40 °C / 93% RH / V_R / 21 days	No visible damage $ \Delta C/C_0 \leq 10\%$ $ \Delta \tan \delta \leq 0.024$ for $C \leq 1 \mu\text{F}$ $ \Delta \tan \delta \leq 0.015$ for $C > 1 \mu\text{F}$ Voltage proof $R_{\text{ins}} \geq 200 \text{ M}\Omega$
Impulse test Endurance	IEC 60384-14:2023	3 impulses $T_b / 1.7 V_R$ / 1000 hours, 1000 V_{RMS} for 0.1 s every hour	No visible damage $ \Delta C/C_0 \leq 10\%$ $ \Delta \tan \delta \leq 0.008$ for $C \leq 1 \mu\text{F}$ $ \Delta \tan \delta \leq 0.005$ for $C > 1 \mu\text{F}$ Voltage proof $R_{\text{ins}} \geq 50\%$ of initial limit
Passive flammability	IEC 60384-14:2023	Flame applied for a period of time depending on capacitor volume	B

Mounting guidelines

1 Soldering

1.1 Solderability of leads

The solderability of terminal leads is tested to IEC 60068-2-20, test Ta, method 1.

Before a solderability test is carried out, terminals are subjected to accelerated ageing (to IEC 60068-2-2, test Ba: 4 h exposure to dry heat at 155 °C). Since the ageing temperature is far higher than the upper category temperature of the capacitors, the terminal wires should be cut off from the capacitor before the ageing procedure to prevent the solderability being impaired by the products of any capacitor decomposition that might occur.

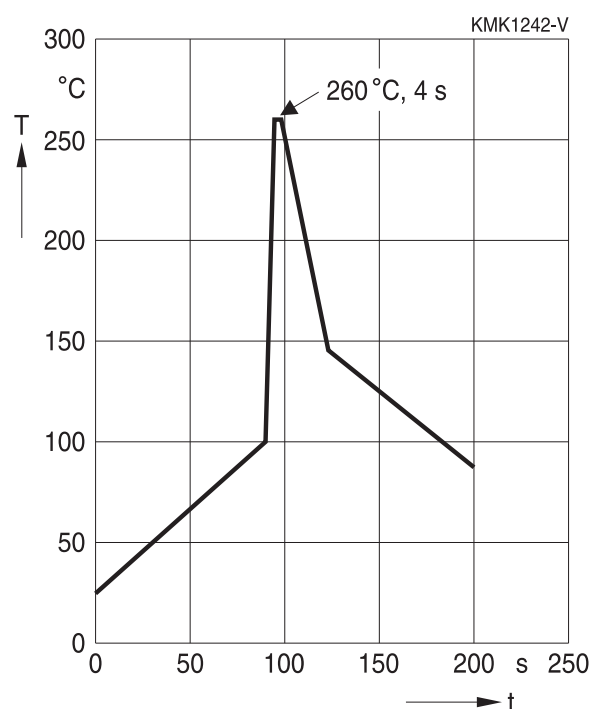
Solder bath temperature	245 ± 3 °C
Soldering time	3.0 ± 0.3 s
Immersion depth	2.0 +0/−0.5 mm from capacitor body or seating plane
Evaluation criteria:	
Visual inspection	Wetting of wire surface by new solder ≥ 95%, free-flowing solder

1.2 Resistance to soldering heat

Resistance to soldering heat is tested to IEC 60068-2-20, test Tb, method 1.

Conditions:

Series	Solder bath temperature	Soldering time
MKT boxed (except $2.5 \times 6.5 \times 7.2$ mm) coated uncoated (lead spacing >10 mm)	260 ± 5 °C	10 ± 1 s
MFP		
MKP (lead spacing >7.5 mm)		
MKT boxed (case $2.5 \times 6.5 \times 7.2$ mm)		5 ± 1 s
MKP (lead spacing ≤ 7.5 mm)		<4 s
MKT uncoated (lead spacing ≤ 10 mm) insulated (B32559)		recommended soldering profile for MKT uncoated (lead spacing ≤ 10 mm) and insulated (B32559)



Immersion depth	2.0 +0/−0.5 mm from capacitor body or seating plane
Shield	Heat-absorbing board, (1.5 ±0.5) mm thick, between capacitor body and liquid solder
Evaluation criteria:	
Visual inspection	No visible damage
$\Delta C/C_0$	2% for MKT/MKP/MFP 5% for EMI suppression capacitors
$\tan \delta$	As specified in sectional specification

1.3 General notes on soldering

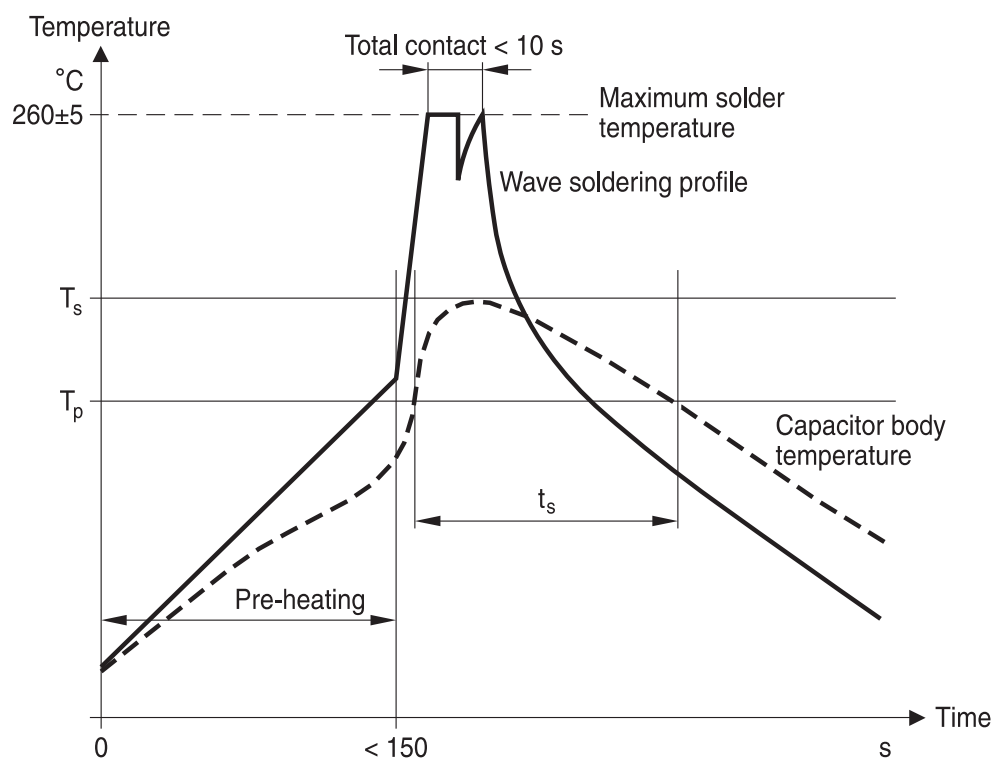
Permissible heat exposure loads on film capacitors are primarily characterized by the upper category temperature T_{max} . Long exposure to temperatures above this type-related temperature limit can lead to changes in the plastic dielectric and thus change irreversibly a capacitor's electrical characteristics. For short exposures (as in practical soldering processes) the heat load (and thus the possible effects on a capacitor) will also depend on other factors like:

- Pre-heating temperature and time
- Forced cooling immediately after soldering
- Terminal characteristics:
diameter, length, thermal resistance, special configurations (e.g. crimping)
- Height of capacitor above solder bath
- Shadowing by neighboring components
- Additional heating due to heat dissipation by neighboring components
- Use of solder-resist coatings

The overheating associated with some of these factors can usually be reduced by suitable countermeasures. For example, if a pre-heating step cannot be avoided, an additional or reinforced cooling process may possibly have to be included.

Recommendations

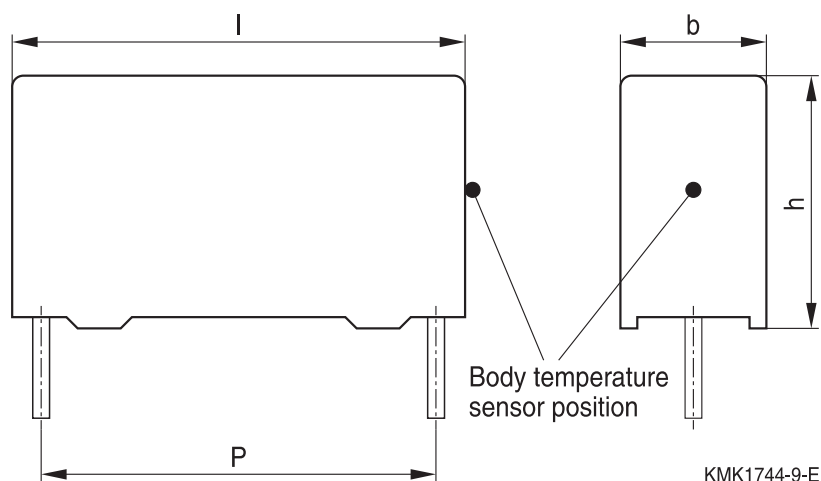
As a reference, the recommended wave soldering profile for our film capacitors is as follows:



T_s : Capacitor body maximum temperature at wave soldering

T_p : Capacitor body maximum temperature at pre-heating

KMK1745-A-E



Body temperature should follow the description below:

- MKP capacitor
 - During pre-heating: $T_p \leq 110\text{ }^{\circ}\text{C}$
 - During soldering: $T_s \leq 120\text{ }^{\circ}\text{C}$, $t_s \leq 45\text{ s}$
- MKT capacitor
 - During pre-heating: $T_p \leq 125\text{ }^{\circ}\text{C}$
 - During soldering: $T_s \leq 160\text{ }^{\circ}\text{C}$, $t_s \leq 45\text{ s}$

When SMD components are used together with leaded ones, the film capacitors should not pass into the SMD adhesive curing oven. The leaded components should be assembled after the SMD curing step.

Leaded film capacitors are not suitable for reflow soldering.

In order to ensure proper conditions for manual or selective soldering, the body temperature of the capacitor (T_s) must be $\leq 120\text{ }^{\circ}\text{C}$.

One recommended condition for manual soldering is that the tip of the soldering iron should be $< 360\text{ }^{\circ}\text{C}$ and the soldering contact time should be no longer than 3 seconds.

For uncoated MKT capacitors with lead spacings $\leq 10\text{ mm}$ (B32560/B32561) the following measures are recommended:

- pre-heating to not more than $110\text{ }^{\circ}\text{C}$ in the preheater phase
- rapid cooling after soldering

Please refer to our Film Capacitors Data Book in case more details are needed.

Cautions and warnings

- Do not exceed the upper category temperature (UCT).
- Do not apply any mechanical stress to the capacitor terminals.
- Avoid any compressive, tensile or flexural stress.
- Do not move the capacitor after it has been soldered to the PC board.
- Do not pick up the PC board by the soldered capacitor.
- Do not place the capacitor on a PC board whose PTH hole spacing differs from the specified lead spacing.
- Do not exceed the specified time or temperature limits during soldering.
- Avoid external energy inputs, such as fire or electricity.
- Avoid overload of the capacitors.
- Consult us if application is with severe temperature and humidity condition.
- There are no serviceable or repairable parts inside the capacitor. Opening the capacitor or any attempts to open or repair the capacitor will void the warranty and liability of TDK Electronics.
- Please note that the standards referred to in this publication may have been revised in the meantime.

The table below summarizes the safety instructions that must always be observed. A detailed description can be found in the relevant sections of the chapters "General technical information" and "Mounting guidelines".

Topic	Safety information	Reference chapter "General technical information"
Storage conditions	Make sure that capacitors are stored within the specified range of time, temperature and humidity conditions.	4.5 "Storage conditions"
Flammability	Avoid external energy, such as fire or electricity (passive flammability), avoid overload of the capacitors (active flammability) and consider the flammability of materials.	5.3 "Flammability"
Resistance to vibration	Do not exceed the tested ability to withstand vibration. The capacitors are tested to IEC 60068-2-6:2007. TDK Electronics offers film capacitors specially designed for operation under more severe vibration regimes such as those found in automotive applications. Consult our catalog "Film Capacitors for Automotive Electronics".	5.2 "Resistance to vibration"

Topic	Safety information	Reference chapter "Mounting guidelines"
Soldering	Do not exceed the specified time or temperature limits during soldering.	1 "Soldering"
Cleaning	Use only suitable solvents for cleaning capacitors.	2 "Cleaning"
Embedding of capacitors in finished assemblies	When embedding finished circuit assemblies in plastic resins, chemical and thermal influences must be taken into account. Caution: Consult us first, if you also wish to embed other uncoated component types!	3 "Embedding of capacitors in finished assemblies"

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Furthermore as we constantly strive to improve our models, the results of tools can change over time and be a non-binding indication only.

Symbols and terms

Symbol	English	German
α	Heat transfer coefficient	Wärmeübergangszahl
α_C	Temperature coefficient of capacitance	Temperaturkoeffizient der Kapazität
A	Capacitor surface area	Kondensatoroberfläche
β_C	Humidity coefficient of capacitance	Feuchtekoeffizient der Kapazität
C	Capacitance	Kapazität
C_R	Rated capacitance	Nennkapazität
ΔC	Absolute capacitance change	Absolute Kapazitätsänderung
$\Delta C/C$	Relative capacitance change (relative deviation of actual value)	Relative Kapazitätsänderung (relative Abweichung vom Ist-Wert)
$\Delta C/C_R$	Capacitance tolerance (relative deviation from rated capacitance)	Kapazitätstoleranz (relative Abweichung vom Nennwert)
dt	Time differential	Differentielle Zeit
Δt	Time interval	Zeitintervall
ΔT	Absolute temperature change (self-heating)	Absolute Temperaturänderung (Selbsterwärmung)
$\Delta \tan \delta$	Absolute change of dissipation factor	Absolute Änderung des Verlustfaktors
ΔV	Absolute voltage change	Absolute Spannungsänderung
dV/dt	Time differential of voltage function (rate of voltage rise)	Differentielle Spannungsänderung (Spannungsflankensteilheit)
$\Delta V/\Delta t$	Voltage change per time interval	Spannungsänderung pro Zeitintervall
E	Activation energy for diffusion	Aktivierungsenergie zur Diffusion
ESL	Self-inductance	Eigeninduktivität
ESR	Equivalent series resistance	Ersatz-Serienwiderstand
f	Frequency	Frequenz
f_1	Frequency limit for reducing permissible AC voltage due to thermal limits	Grenzfrequenz für thermisch bedingte Reduzierung der zulässigen Wechselspannung
f_2	Frequency limit for reducing permissible AC voltage due to current limit	Grenzfrequenz für strombedingte Reduzierung der zulässigen Wechselspannung
f_r	Resonant frequency	Resonanzfrequenz
F_D	Thermal acceleration factor for diffusion	Therm. Beschleunigungsfaktor zur Diffusion
F_T	Derating factor	Deratingfaktor
i	Current (peak)	Stromspitze
I_C	Category current (max. continuous current)	Kategoriestrom (max. Dauerstrom)

Symbol	English	German
I_{RMS}	(Sinusoidal) alternating current, root-mean-square value	(Sinusförmiger) Wechselstrom
i_z	Capacitance drift	Inkonstanz der Kapazität
k_0	Pulse characteristic	Impulskennwert
L_S	Series inductance	Serieninduktivität
λ	Failure rate	Ausfallrate
λ_0	Constant failure rate during useful service life	Konstante Ausfallrate in der Nutzungsphase
λ_{test}	Failure rate, determined by tests	Experimentell ermittelte Ausfallrate
P_{diss}	Dissipated power	Abgegebene Verlustleistung
P_{gen}	Generated power	Erzeugte Verlustleistung
Q	Heat energy	Wärmeenergie
ρ	Density of water vapor in air	Dichte von Wasserdampf in Luft
R	Universal molar constant for gases	Allg. Molarkonstante für Gas
R	Ohmic resistance of discharge circuit	Ohmscher Widerstand des Entladekreises
R_i	Internal resistance	Innenwiderstand
R_{ins}	Insulation resistance	Isolationswiderstand
R_P	Parallel resistance	Parallelwiderstand
R_S	Series resistance	Serienwiderstand
S	severity (humidity test)	Schärfegrad (Feuchtetest)
t	Time	Zeit
T	Temperature	Temperatur
τ	Time constant	Zeitkonstante
$\tan \delta$	Dissipation factor	Verlustfaktor
$\tan \delta_D$	Dielectric component of dissipation factor	Dielektrischer Anteil des Verlustfaktors
$\tan \delta_P$	Parallel component of dissipation factor	Parallelanteil des Verlustfaktors
$\tan \delta_S$	Series component of dissipation factor	Serienanteil des Verlustfaktors
T_A	Temperature of the air surrounding the component	Temperatur der Luft, die das Bauteil umgibt
T_{max}	Upper category temperature	Obere Kategorietemperatur
T_{min}	Lower category temperature	Untere Kategorietemperatur
t_{OL}	Operating life at operating temperature and voltage	Betriebszeit bei Betriebstemperatur und -spannung
T_{op}	Operating temperature, $T_A + \Delta T$	Betriebstemperatur, $T_A + \Delta T$
T_R	Rated temperature	Nenntemperatur
T_{ref}	Reference temperature	Referenztemperatur
t_{SL}	Reference service life	Referenz-Lebensdauer

Symbol	English	German
V_{AC}	AC voltage	Wechselspannung
V_C	Category voltage	Kategoriespannung
$V_{C,RMS}$	Category AC voltage	(Sinusförmige) Kategorie-Wechselspannung
V_{CD}	Corona-discharge onset voltage	Teilentlade-Einsatzspannung
V_{ch}	Charging voltage	Ladespannung
V_{DC}	DC voltage	Gleichspannung
V_{FB}	Fly-back capacitor voltage	Spannung (Flyback)
V_i	Input voltage	Eingangsspannung
V_o	Output voltage	Ausgangsspannung
V_{op}	Operating voltage	Betriebsspannung
V_p	Peak pulse voltage	Impuls-Spitzenspannung
V_{pp}	Peak-to-peak voltage Impedance	Spannungshub
V_R	Rated voltage	Nennspannung
$\hat{V}_R$	Amplitude of rated AC voltage	Amplitude der Nenn-Wechselspannung
V_{RMS}	(Sinusoidal) alternating voltage, root-mean-square value	(Sinusförmige) Wechselspannung
V_{SC}	S-correction voltage	Spannung bei Anwendung "S-correction"
V_{sn}	Snubber capacitor voltage	Spannung bei Anwendung "Beschaltung"
Z	Impedance	Scheinwiderstand
e	Lead spacing	Rastermaß

Important notes

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1. Some parts of this publication contain **statements about the suitability of our products for certain areas of application**. These statements are based on our knowledge of typical requirements that are often placed on our products in the areas of application concerned. We nevertheless expressly point out **that such statements cannot be regarded as binding statements about the suitability of our products for a particular customer application**. As a rule we are either unfamiliar with individual customer applications or less familiar with them than the customers themselves. For these reasons, it is always ultimately incumbent on the customer to check and decide whether a product with the properties described in the product specification is suitable for use in a particular customer application.
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Important notes

8. The trade names EPCOS, CarXield, CeraCharge, CeraDiode, CeraLink, CeraPad, CeraPlas, CSMP, CTVS, DeltaCap, DigiSiMic, FilterCap, FormFit, InsuGate, LeaXield, MediPlas, MiniBlue, MiniCell, MKD, MKK, ModCap, MotorCap, PCC, PhaseCap, PhaseCube, PhaseMod, PhiCap, PiezoBrush, PlasmaBrush, PowerHap, PQSine, PQvar, SIFERRIT, SIFI, SIKOREL, SilverCap, SIMDAD, SiMic, SIMID, SineFormer, SIOV, SurfIND, ThermoFuse, WindCap, XieldCap are **trademarks registered or pending** in Europe and in other countries. Further information will be found on the Internet at www.tdk-electronics.tdk.com/trademarks.

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